

POC

Jira → GitLab

Sync Service

Automated replication of Jira Stories, Bugs & Tasks
to GitLab milestones using a .NET 8 Windows Service

Problem Statement

Three challenges driving the need for this sync service

1

Limited Jira Access — Entire Team Cannot See Work Items

Stories and Bugs are created and managed in Jira, but only a few team members have Jira licences. Developers, QA, and other contributors who work in GitLab daily have no visibility into planned work items, sprint scope, or ticket details — creating a blind spot across the team.

2

Disconnected Tools — Code in GitLab, Work Tracking in Jira

All code repositories live in GitLab which every team member can access, but sprint planning, stories, and bugs remain siloed in Jira. Developers must context-switch between two systems to understand what to build, making it difficult to track sprint progress or link commits to work items.

3

Manual Work Tracking — Team Managing Tasks & Status in Excel

Without access to Jira or a shared system, team members resort to maintaining tasks, assignments, and work status in Excel spreadsheets. This leads to version conflicts, stale data, no sprint-wise progress tracking, and zero traceability between work items and the actual code being delivered.

Solution Benefits

What the team gains by syncing Jira to GitLab



Whole Team Visibility

Every team member — developer, QA, designer — can view sprint stories, bugs and tasks directly in GitLab without needing a Jira licence.



Single Source of Truth

Replaces fragmented Excel sheets with structured GitLab issues. Work status, assignees and story points are always up to date and version-controlled.



Sprint-wise Progress Tracking

Tickets are mapped to GitLab milestones matching the current sprint. Teams can track velocity, completion rate and sprint health directly in GitLab.



Code ↔ Work Item Traceability

With issues in GitLab, developers can reference ticket IDs in commits and merge requests — linking code changes directly to the business requirement.



Zero Manual Effort

The service runs automatically every 15 minutes. No copy-pasting, no spreadsheet updates — the sync is fully hands-free once deployed.



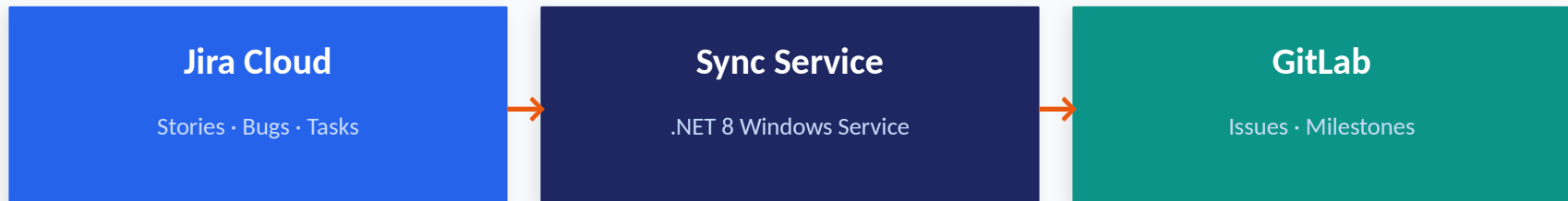
Better Reporting & Retrospectives

GitLab burndown charts, issue boards and milestone reports give the team accurate sprint data for retrospectives and stakeholder updates.

✓ No Jira licence needed for developers ✓ Replaces Excel tracking ✓ Sprint progress in GitLab ✓ Fully automated

Solution Overview

A .NET 8 Windows Service that automatically replicates Jira tickets to GitLab on a configurable schedule



Scheduled Sync

Runs every 15 min (configurable). Auto-retry on failure.



Smart Routing

Active milestone today → sprint. No milestone → Backlog.



Create & Update

New tickets are created; existing ones are updated in place.



Field Mapping

ID, Description, Assignee, Dates, Story Points all synced.



Full Logging

Microsoft ILogger with Debug/Info/Warn/Error levels.

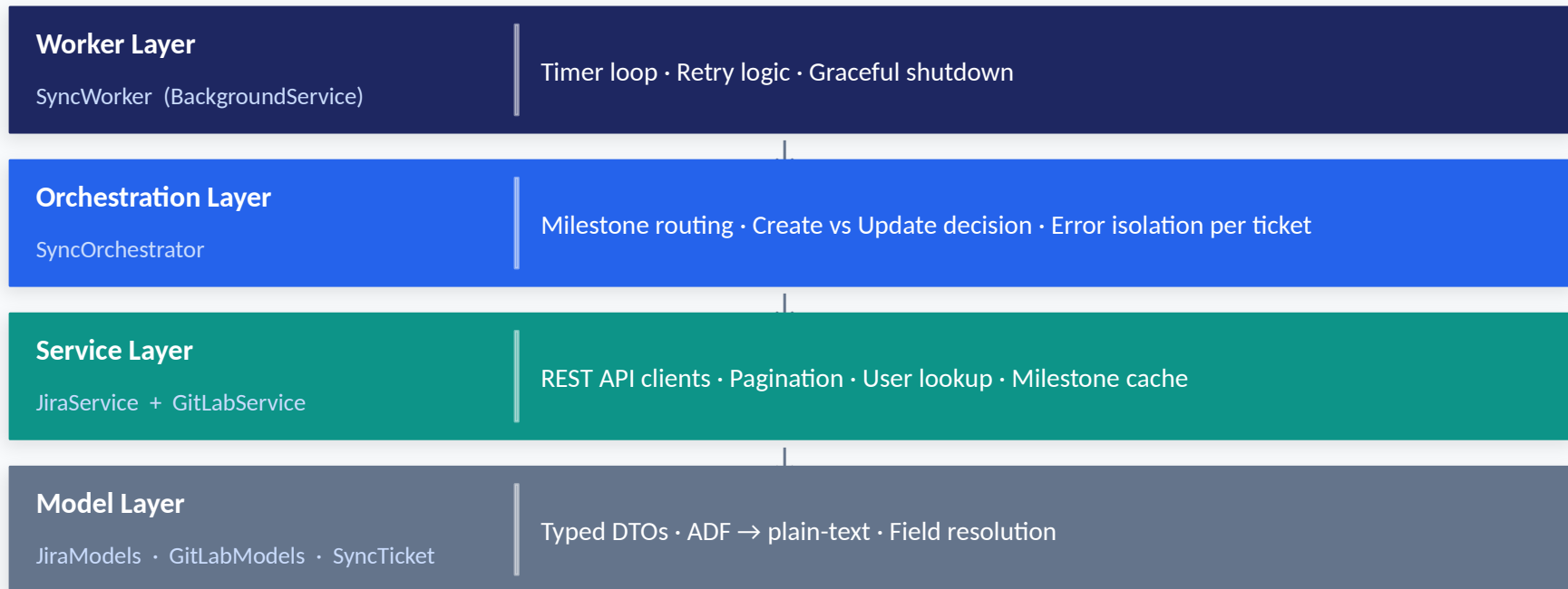


Config-Driven

All settings in appsettings.json — no recompile needed.

Architecture

Layered .NET 8 service with clean separation of concerns



Field Mapping

How Jira ticket data maps to GitLab issue fields

Jira Field	GitLab Field	Notes
Story / Bug / Task Key	Issue title prefix [PROJ-42]	Enables duplicate detection on re-sync
Summary	Issue title (after prefix)	Full summary preserved
Description (ADF)	Issue description (Markdown)	Atlassian ADF flattened to plain text
Assignee display name	Assignee (matched by name/email)	Skipped on free-tier GitLab (403)
Sprint start date → end date	Shown in description: From → To	Also drives milestone selection
Story Points (customfield)	Issue weight	Tries 3 common custom field names
Issue type	GitLab label (story / bug / task)	Configurable via LabelMappings